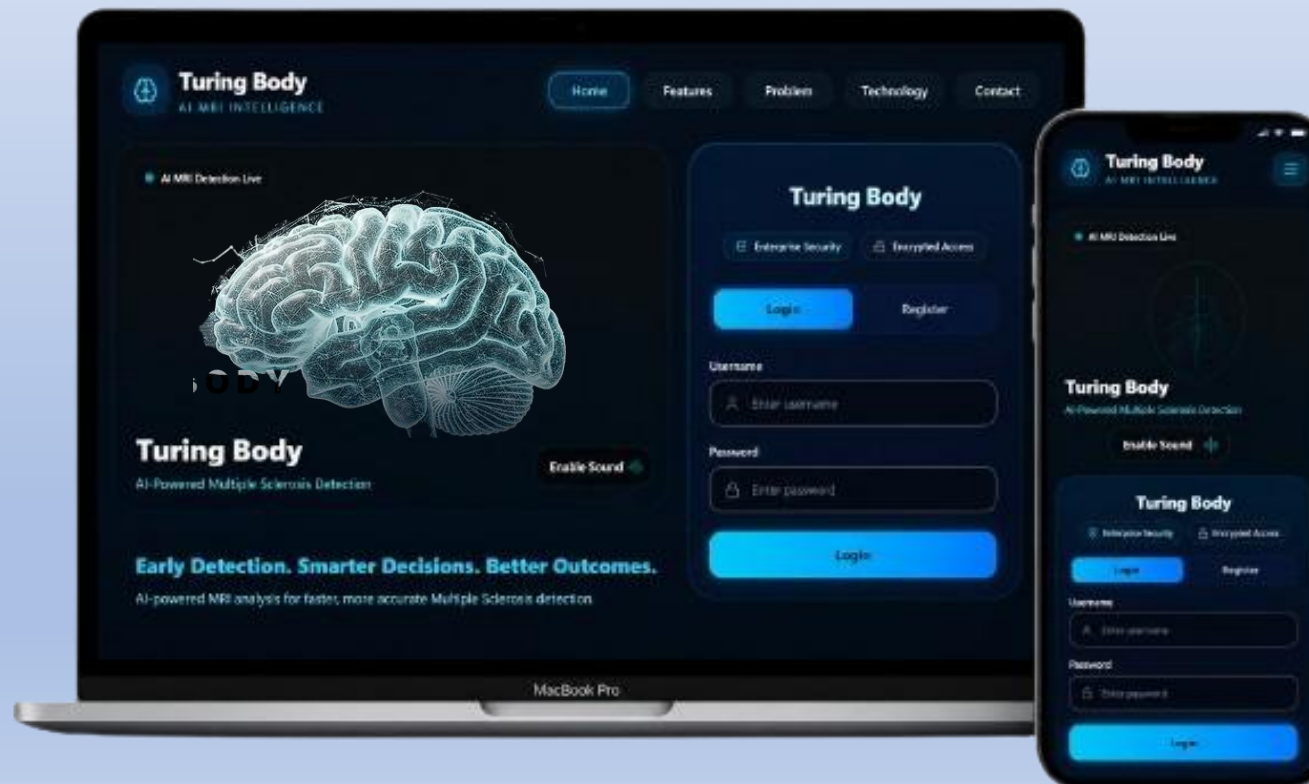


Turing Body

AI-Based Early Detection of Multiple Sclerosis Using MRI Image Analysis



Introduction of Turing Body

❑ Turing Body is an AI-powered medical imaging platform developed for the automated detection and segmentation of Multiple Sclerosis (MS) lesions from MRI brain scans.

The system combines:

- Deep Learning
- Medical Image Processing
- Spring Boot Backend APIs
- React Frontend Framework

❑ The core intelligence of the system is based on the U-Net Deep Learning architecture, which performs pixel-level lesion segmentation from MRI FLAIR images. The platform allows doctors to upload MRI scans, automatically analyze lesion regions, estimate disease severity, visualize segmentation results, and generate PDF-based medical reports.

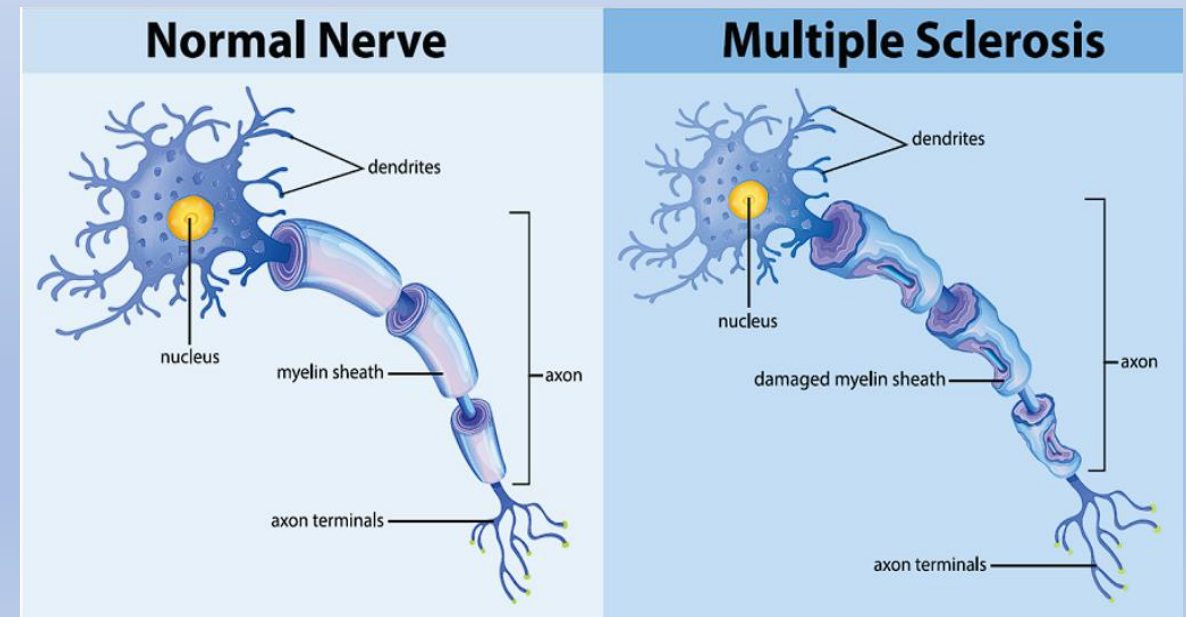
Turing Body was designed to improve:

- lesion analysis efficiency,
- segmentation automation,
- medical visualization,
- and AI-assisted diagnosis workflows.

Problem Statement

- Multiple Sclerosis (MS) is a chronic neurological disorder that affects the central nervous system by damaging the myelin sheath — the protective covering surrounding nerve fibers in the brain and spinal cord.
- MRI (Magnetic Resonance Imaging) is one of the primary diagnostic tools used to identify MS lesions. However, manual MRI lesion analysis presents several challenges:
 - Time-consuming examination process
 - Difficulty in identifying small and irregular lesions
 - Dependency on radiologist expertise
 - Risk of human error and inconsistent interpretation
 - Challenging analysis for large MRI datasets

Therefore, there is a need for an intelligent automated system that can accurately detect, segment, and analyze MS lesions from MRI brain scans using Artificial Intelligence and Deep Learning techniques.



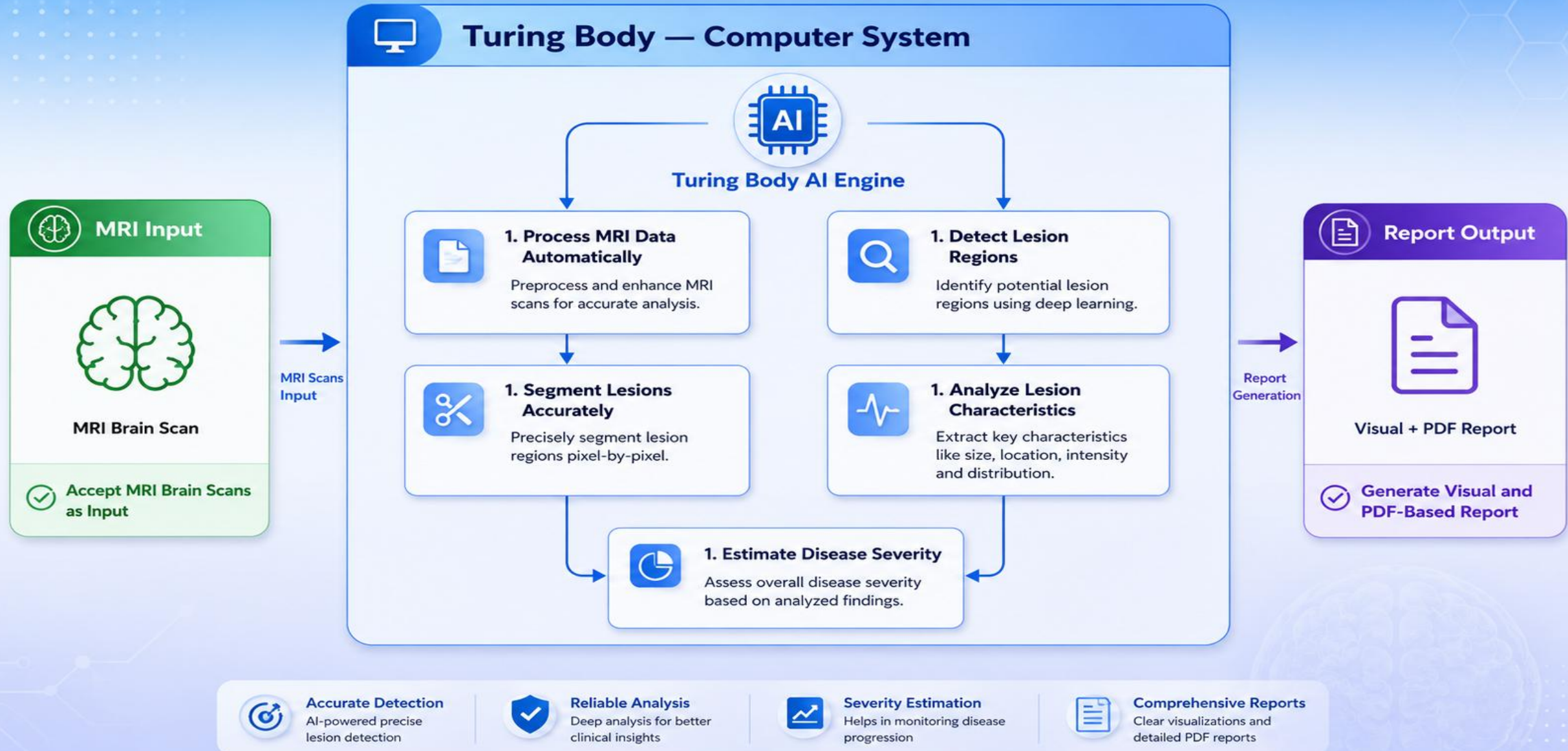
Existing System Limitations

Traditional MRI Analysis	Traditional Machine Learning
▪ Requires manual MRI inspection ▪ Time-consuming process ▪ Depends on radiologist expertise ▪ Inconsistent lesion interpretation ▪ Difficult for large MRI datasets	▪ Requires manual feature extraction ▪ Poor pixel-wise segmentation ▪ Limited lesion localization ▪ Lower segmentation accuracy ▪ Cannot detect fine lesion boundaries

Result

Traditional approaches fail to provide fast, accurate, and automated MRI lesion segmentation, making MS diagnosis more challenging and time-intensive.

Objective of Turing Body



Dataset & Preprocessing Pipeline



DATASET USED

The project utilizes MRI FLAIR brain scans along with corresponding ground truth lesion masks for supervised Deep Learning-based lesion segmentation.



Dataset Characteristics



Dataset Type: MRI FLAIR Brain Scans



Ground Truth: Expert-annotated lesion masks



File Format: .nii (NIfTI Medical Imaging Format)



DATASET CONTAINS

- ✓ Original MRI brain scan volumes
- ✓ Expert-labeled lesion segmentation masks
- ✓ The annotated lesion masks serve as ground truth labels for training the U-Net segmentation model.



DATA PREPROCESSING PIPELINE

To prepare MRI data for Deep Learning segmentation, a preprocessing pipeline was implemented to standardize and optimize the input data.



PREPROCESSING STEPS:

01



MRI File Reading

02



Slice Extraction from 3D MRI Volumes

03



Intensity Normalization

04



Image Resizing to 128 × 128

05



Lesion Mask Preparation

06



Train-Test Dataset Splitting

Core Model U-Net Architecture

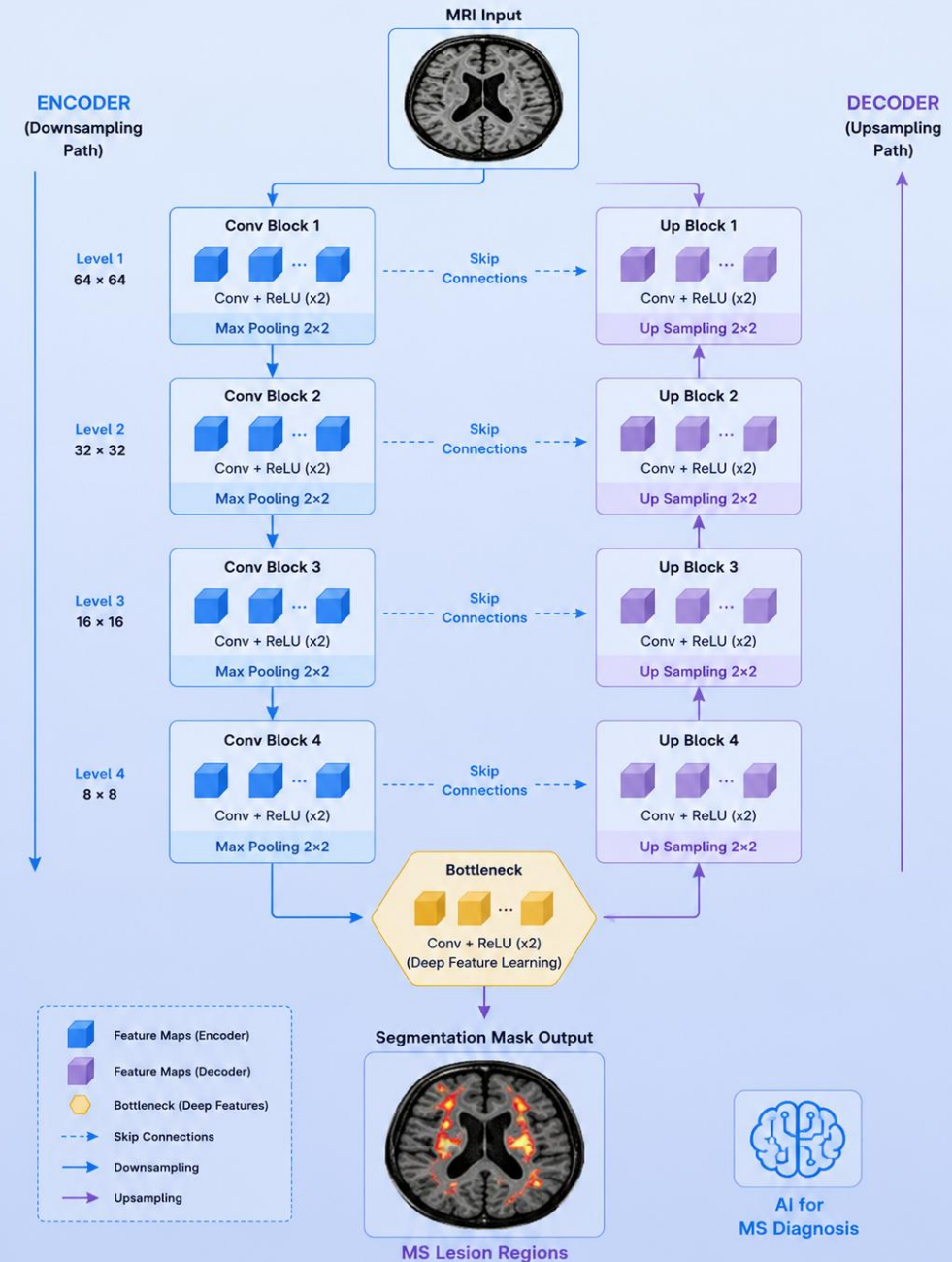
U-Net Convolutional Neural Network

- The proposed system uses a **U-Net Deep Learning architecture** for automated Multiple Sclerosis lesion segmentation from MRI brain scans.
- U-Net is a specialized Convolutional Neural Network (CNN) designed specifically for **biomedical image segmentation**, where precise pixel-level localization is required.

Why U-Net Was Selected

- Specifically designed for medical image segmentation
- Performs effectively on small medical datasets
- Provides accurate lesion localization
- Uses skip connections for fine feature recovery
- Produces pixel-level segmentation predictions
- Widely used in biomedical imaging research
- Preserves important spatial information

U-Net Architecture for Multiple Sclerosis MRI Lesion Segmentation



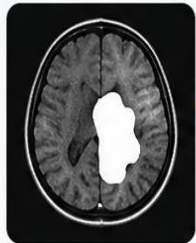
Model Performance & Metrics



EVALUATION METRICS for Lesion Segmentation

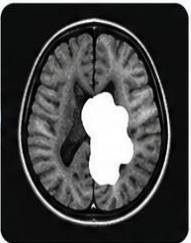
WHAT IS BEING EVALUATED?

Predicted
Lesion Mask



VS

Ground Truth
Lesion Mask



OVERLAP



Higher overlap
= **Better** segmentation
accuracy

LEGEND

- Predicted Lesion
- Ground Truth Lesion
- Overlap (Intersection)



These metrics help **quantify** how well the predicted lesions match the ground truth, guiding the model to learn **better** and segment more accurately.



COMBINATION OF THESE METRICS leads to accurate, reliable, and robust lesion segmentation models.

RIGHT SIDE — EVALUATION METRICS

1



DICE COEFFICIENT

- Measures overlap between:
 - predicted** lesion mask
 - ground truth** mask
- Higher Dice = **Better** segmentation accuracy

2



DICE LOSS

- Minimizes segmentation error
- Improves lesion overlap quality
- Helps the model learn better boundaries and shapes

3



BINARY CROSS ENTROPY (BCE)

- Measures pixel-wise classification error between:
 - lesion pixels**
 - non-lesion pixels**
- Improves segmentation **precision**



Validation Performance

Validation Dice Score

56%



Measures segmentation
quality on unseen MRI
scans.



Training Performance

Training Dice Score

71%



Indicates effective
lesion feature
learning.



Performance Analysis



Stable lesion segmentation on unseen MRI scans



Effective feature learning using U-Net



Reasonable generalization for limited medical datasets



Improved lesion localization through Deep Learning



TECHNOLOGY STACK – TURING BODY

AI-Powered MRI Lesion Segmentation | Secure Backend | Modern Web Interface | Scalable Deployment



MACHINE LEARNING & AI

Technology	Purpose
 Python	ML Development
 TensorFlow / Keras	Deep Learning Framework
 U-Net	MRI Lesion Segmentation
 OpenCV	Image Processing
 NumPy	Numerical Computation
 SciPy	Lesion Analysis
 nibabel	MRI .nii File Handling



BACKEND TECHNOLOGIES

Technology	Purpose
 Spring Boot	Backend Framework
 Spring Security	Authentication & Authorization
 JWT	Secure Token Authentication
 Spring Data JPA	Database ORM
 MongoDB	NoSQL Database
 Maven	Dependency Management
 Docker	Containerization



FRONTEND TECHNOLOGIES

Technology	Purpose
 Next.js	Frontend Framework
 React.js	UI Development
 Tailwind CSS	Responsive Styling
 Axios / Fetch API	API Communication
 TypeScript / JavaScript	Frontend Logic
 Vercel	Frontend Deployment



DEPLOYMENT & TOOLS

Tool / Platform	Purpose
 	Version Control
 Google Colab	Model Training
 VS Code	Development Environment
 Postman	API Testing
 render	Backend Deployment
 Render / Railway	Backend Deployment



POWERING AN END-TO-END AI SYSTEM



AI models segment lesions from MRI scans



Secure backend handles data & users

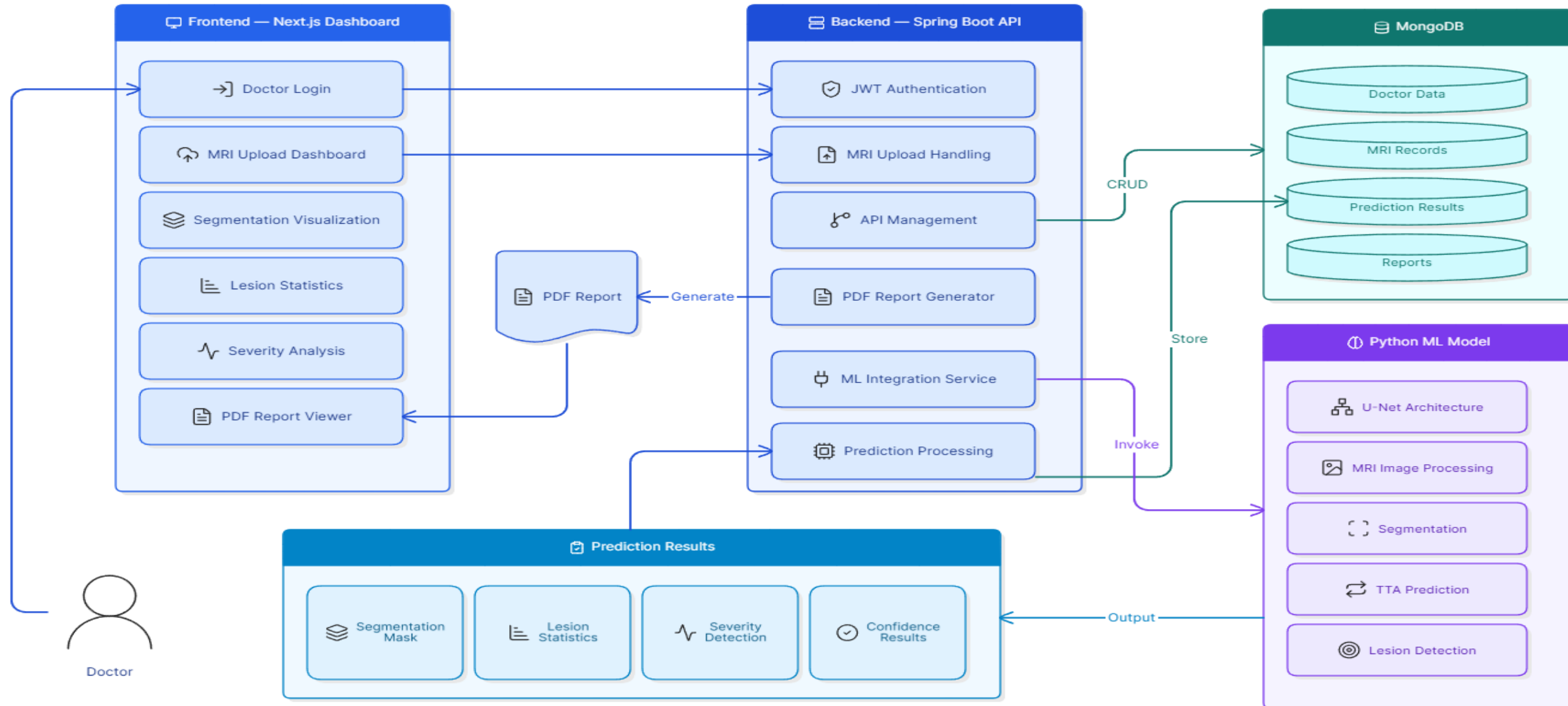


Modern frontend delivers seamless UX



Deployed & scalable for real-world use

System Architecture of Turing Body



Background Study(Research Papers studied)

Sr. number	Paper Title	Publisher	Year	Take-away points
1	Deep Learning-Based Multiple Sclerosis Lesion Segmentation in MRI	IEEE	2021	CNN models improve detection of MS lesions in brain MRI scans
2	Automated Detection of Multiple Sclerosis Using Convolutional Neural Networks	Springer	2020	Deep learning provides higher accuracy than traditional ML methods
3	MRI Brain Image Classification for Neurological Disorders Using AI	Elsevier	2022	AI enables early and consistent diagnosis of neurological diseases
4	Medical Image Analysis Using Deep Learning Techniques	IEEE	2023	Deep learning enhances feature extraction and classification performance

Existing Research vs Proposed System

System	Accuracy / Performance	Limitations
Traditional Manual MRI Analysis	Depends on radiologist expertise	Time-consuming and inconsistent
Classical Machine Learning Approaches	Moderate classification accuracy	Requires manual feature extraction
Existing CNN-Based Systems (2022–2023)	~70–90% classification accuracy	Mostly classification only
Medical Image Analysis using Deep Learning	~55.93% Dice Segmentation Accuracy	Limited lesion boundary localization
Proposed System – Turing Body (2025)	Automated semantic segmentation using U-Net	Improved lesion localization, severity analysis, and automated reporting

References

➤ O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional Networks for Biomedical Image Segmentation,” Proceedings of MICCAI, pp. 234–241, 2015.	<u>Unet research paper</u>
➤ K. He, X. Zhang, S. Ren, and J. Sun, “Deep Residual Learning for Image Recognition,” Proceedings of CVPR, pp. 770–778, 2016.	<u>ResNet Research Paper</u>
➤ I. Goodfellow, Y. Bengio, and A. Courville, <i>Deep Learning</i> , MIT Press, 2016.	<u>Deep Learning Book</u>
➤ D. P. Kingma and J. Ba, “Adam: A Method for Stochastic Optimization,” Proceedings of ICLR, 2015.	<u>Adam Optimizer Paper</u>
➤ F. Chollet, <i>Deep Learning with Python</i> , Manning Publications, 2018.	<u>Deep Learning with Python Book</u>
➤ A. Krizhevsky, I. Sutskever, and G. E. Hinton, “ImageNet Classification with Deep Convolutional Neural Networks,” Proceedings of NIPS, pp. 1097–1105, 2012.	<u>AlexNet Research Paper</u>

Project View



Live project Link : <https://ai-ms-detection-system.vercel.app/>

Turing Body

AI MRI INTELLIGENCE

Home

Features

Problem

Technology

Contact

AI MRI Detection Live

Turing Body

AI-Powered Multiple Sclerosis Detection

Early Detection. Smarter Decisions. Better Outcomes.

Enterprise Security

Encrypted Access

Login

Register

Username

Enter username

Password

Enter password

Login

TURING BODY

Dashboard

Patients

Upload MRI

Results

History

Analytics

Settings

AI Prediction Results

MRI Analysis & Clinical Diagnostic Report

Patient ID: 6a07dd68a0edb90658925caa

Send Report

PDF Report

MRI Scan Preview

AI generated lesion overlay analysis

Mask Image

Overlay Image

AI Prediction Analysis

Prediction

Lesion Volume

Severity

Lesion Detected

320

Moderate MS

Report Summary

AI analysis detected 1 lesion(s) primarily located in the Periventricular region. The detected lesions show an estimated total lesion volume of 320 mm³ with the largest lesion area measuring 80 pixels. Disease severity is currently classified as Moderate MS. Average lesion size is approximately 80 pixels, suggesting ongoing white matter abnormality patterns associated with Multiple Sclerosis. Further neurological and radiological evaluation is recommended for clinical confirmation and treatment planning.

AI MRI Lesion Detection System

Developed by Alexian Keshiyaj

Multiple Sclerosis MRI Analysis Report

Report Information

Patient ID

uploaded_scan

MRI Slices Analyzed

1

AI Lesion Detection Summary

Lesion Slices Detected

1

Total Lesion Pixels

80

Estimated Lesion Volume (mm³)

320

Disease Severity

Moderate MS

MRI Lesion Detection Visualization

Input MRI

Detected Lesion Mask

Mask Overlay on MRI

III. Lesion Statistics

Total Lesions Detected

1

Average Lesion Size (pixels)

80.00

Largest Lesion Area (pixels)

11475

Primary Brain Region

Periventricular

Research & Clinical Report

AI Assisted Diagnosis Tool

Generated using Deep Learning MRI Lesion Segmentation Model

Page 1 of 1

TURING BODY

Dashboard

Quick Actions

Recent Activity

Dr. Anvita

Dashboard

Patients

Upload MRI

Results

History

Analytics

Settings

Total Patients

128

MRI Uploads

342

Weekly MRI Uploads

Mon

Tue

Wed

Thu

Fri

Sat

Positive Cases

89

Prediction Accuracy

95%

Quick Actions

Add Patient

Upload MRI

Generate Report

System Status

Recent Activity

MRI uploaded for Patient #1023

TURING BODY

Dashboard

Patients

Upload MRI

Results

History

Analytics

Settings

AI MRI Upload System

Upload MRI scans and run AI-powered MS detection analysis.

MRI Scan Upload

Supported: JPG, PNG, DICOM

Drag & Drop MRI Scan

Upload brain MRI scans for AI-powered Multiple Sclerosis detection and analysis.

Browse MRI Files

Run AI Prediction

Patient Clinical Details

Patient ID

Checkup

Patient ID

Done

Weight (kg)

Height (cm)

Weight

Height

Clinical Notes

Doctor observations and clinical notes...

TURING BODY

Dashboard

Patients

Upload MRI

Results

History

Analytics

Settings

Analytics & Reports

AI-powered MRI detection analytics and prediction insights.

Total MRI Scans

1,284

+12% this month

High-Risk Cases

342

Critical detections

Medium Risk Cases

511

Moderate detections

Low Risk Cases

431

Low-risk predictions

MRI Prediction Trends

Jan

Feb

Mar

Apr

May

Jun

AI Insights

High-risk lesions increased by 12%

Most affected age group: 30-45

AI confidence improved to 96%

MRI detection latency reduced by 18%

22-May-26

Project Presentation 2026

14

CHALLENGES & SOLUTIONS

Overcoming Roadblocks. Building Reliable Solutions.



Conclusion

- ❑ Turing Body successfully demonstrates the integration of Deep Learning, medical image processing, and full-stack web technologies for automated Multiple Sclerosis lesion detection and segmentation from MRI brain scans.
- ❑ The system utilizes a U-Net-based segmentation model to accurately identify lesion regions, analyze lesion characteristics, estimate disease severity, and generate visual PDF-based reports.
- ❑ By combining a Next.js frontend, Spring Boot backend, MongoDB database, and Python-based ML pipeline, the project provides a complete AI-powered healthcare platform that improves automation, medical visualization, and MRI analysis efficiency.
- ❑ The project highlights the practical application of Artificial Intelligence in medical imaging and showcases the potential of Deep Learning in assisting modern healthcare diagnosis workflows.

Thank you